Deep Learning Coursework Guidance

*Note: Deep Learning Final Project Report

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***Abstract*—This course work focuses on two tasks: shape classification and contact force estimation from images of a GelSight vision-based tactile sensor.**

***Index Terms*—Deep Learning Coursework**

I. Introduction

A. Background

Tactile sensing plays a crucial role in robotics, prosthetics, and human–robot interaction, where a robot must perceive the objects it touches and the forces it applies. Vision-based tactile sensors such as GelSight convert contact into images: an illuminated elastomer gel deforms when an object presses into it, and an internal camera records the change in shading and colour. Some sensors print a grid of black markers on the gel, whose displacement reveals how the gel is stretched by tangential (shear) loading.

Tactile image datasets used for tasks like shape classification and force estimation introduce unique challenges. For instance, the contact region often occupies only a small part of an otherwise uniform image, and shallow contacts produce subtle, low-contrast deformations. Moreover, the images are strongly influenced by the gel, the illumination, and the camera of each individual sensor.

Potential Challenges:

- Subtle and spatially small contact cues, which can be lost when images are downsampled.
- Coupling between normal and tangential force, with tangential forces that are small compared to the normal force in many samples.
- Scarcity of labelled data due to the reliance on robot-controlled data collection and force/torque sensors.

Two tasks for coursework:

1) Shape Classification: Shape classification identifies the geometry of the object in contact with the sensor (10 shape classes). In robotic manipulation, it is vital for object recognition when vision is occluded by the gripper or the object itself.

Recent advances in deep learning, particularly CNNs, enable automatic feature learning from raw tactile images, significantly improving shape classification.

2) Force Estimation: Force estimation predicts the 3D contact force vector from a single tactile image: the normal force f_z and the tangential (shear) forces f_x and f_y. This is critical for grasp control, slip detection, and safe physical interaction.

Note: Integrating shape classification and force estimation into a unified framework enhances the perception capabilities of robotic hands.

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II. Tasks and Methodology

A. Task 1: Data Preparation and Preprocessing

Objectives:

1) Understand the dataset structure.
2) Prepare data for input into deep learning models.

Steps:

1) **Data Loading:** Write scripts to read images from the `images/` folders and parse the `labels.csv` files. For shape classification, `labels.csv` links each image to a `class_name` and `class_id`. For force estimation, it links each image to the force values `fx`, `fy`, `fz` (N), a `contact_id`, and an `is_shear` flag (normal press or shear).
2) **Exploratory Data Analysis:** Visualize sample images and analyze the class and force distributions, including the difference between normal-press and shear images.
3) **Augmentation:** Optional. Justify that each augmentation is physically valid (e.g., flips and rotations change the direction of f_x, f_y).
4) **Normalization:** Normalize pixel intensities and standardize force values.
5) **Data Splitting:** Split data into 60% training, 20% validation, and 20% testing sets, ensuring balanced class distribution for shape classification and grouping images by `contact_id` for force estimation.

B. Task 2: Shape Classification

Objectives:

1) Design and implement a deep learning model for shape classification.
2) Evaluate performance using classification metrics.

Steps:

1) **Model Design:** Implement and compare at least three CNN architectures.
2) **Model Optimization and Regulation:** Optional

3) **Evaluation:** Measure Accuracy, Precision, Recall, F1 Score, and Confusion Matrix.
4) **Error Analysis:** Identify patterns in misclassified samples.

C. Task 3: Force Estimation (Regression)

Objectives:

1) Build a regression model for 3D contact force estimation.
2) Achieve accurate estimation of both the normal and the tangential force.

Steps:

1) **Model Design:** Output three continuous values (f_x, f_y, f_z) for force prediction.
2) **Advanced Training Strategy:** (Optional)
3) **Evaluation:** Measure RMSE per axis and overall, and compare results across architectures.
4) **Tangential Force Analysis:** Report the tangential error separately for normal-press and shear images, and compare with a mean-prediction baseline (see the dataset README).
5) **Error Analysis:** Visualize prediction errors and discuss edge cases.

D. Task 4: Hyperparameter Tuning (Optional)

Objectives:

1) Optimize performance through systematic hyperparameter tuning.
2) Justify selected hyperparameters based on experiments.

E. Task 5: Evaluation and Reporting

Objectives:

1) Present results with clarity and depth.
2) Provide insights into performance, limitations, and future directions.

Steps:

1) **Test Set Evaluation:** We will test your model performance in our hidden test set. Your `predict.py` must follow the submission interface in the dataset README.
2) **Final Report:** Prepare an IEEE-formatted document with sections including Abstract, Introduction, Objectives, Methodology, Experiments, Evaluation, Conclusions, and Future Work.

III. Deliverables (Submissions)

1) **Code:** Scripts for data preprocessing, model training, evaluation, and tuning.
2) **Models:** Models for testing. Include an inference script `predict.py` that follows the submission interface. A template is provided in the `example_submission/` folder of the dataset.
3) **Report:** IEEE-formatted double-column document (3–6 pages) with results and insights.